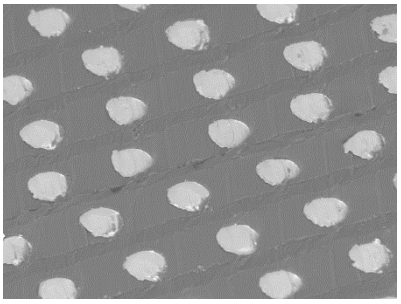
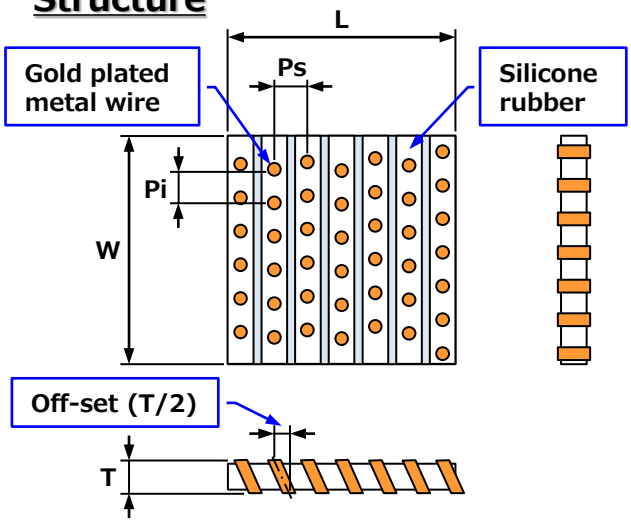
Shin-Etsu Inter-Connector "MT" series

- This sheet type connector for inspection is anisotropic conductive sheet type, which consists of gold plated metal wires embedded into silicone rubber. The conductive wires are designed in fine pitched matrix array.





Structure



Features

- Excellent RF characteristic
- Wide Range of Pitch and Sizes
- Accommodates various device types
- Ease of Repair and Replacement

Dimensions

Type	Pi(P) (mm)	Ps (mm)	L (mm)	W (mm)	T (mm)	Off-set (mm)	Hardness	Protrusion of Wire (mm)
MT	0.1	0.1	5 to 50	5 to 50	0.5 1.0 2.0	0.125 0.5 1.0	30	less than 0.01
MT-2X	0.075	0.075						
MT-4X	0.05	0.05						
MT-6X	0.032	0.05						

- *There is also the "MT-E series" with the above specifications, in which the metal wire protrusion is 0.035mm.*

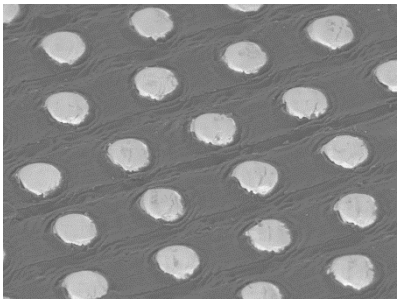
Operating Precautions

- In the case of solder electrodes, solder debris become deposited on the contact area after repeated use. Please clean the surface with a toothbrush; and remove the debris. Otherwise, resistance value will increase due to oxidation. When the solder debris is removed, the resistance value tends to recover.

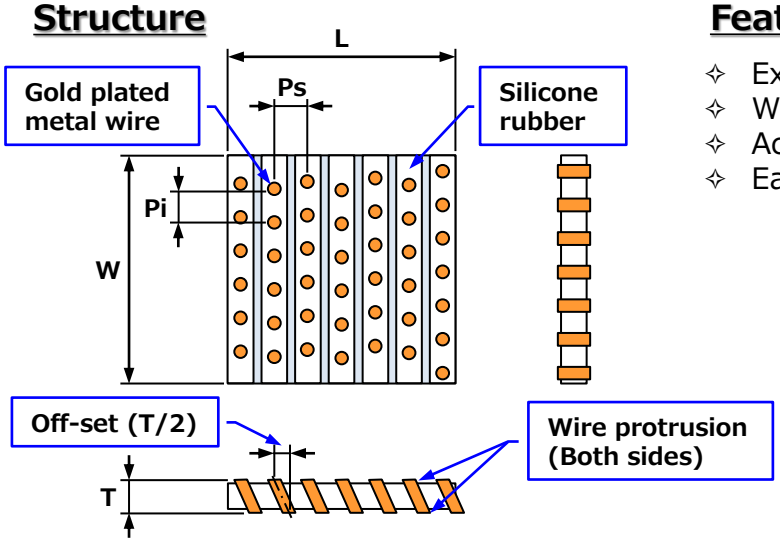
Shin-Etsu Inter-Connector "MT-T" series

- This sheet type connector for inspection is anisotropic conductive sheet type, which consists of gold plated metal wires embedded into silicone rubber. The conductive wires are designed in fine pitched matrix array.
- MT-"T" Type is a thin version of the MT series, and the conductor is short and has excellent high frequency transmission characteristics.





Structure



Features

- Excellent RF characteristic
- Wide Range of Pitch and Sizes
- Accommodates various device types
- Ease of Repair and Replacement

Dimensions

Type	Pi(P) (mm)	Ps (mm)	L (mm)	W (mm)	T (mm)	Off-set (mm)	Hardness	Protrusion of Wire (mm)
MT-T	0.1	0.1	5 to 50	5 to 50	(0.1) 0.15 0.25	(0.05) 0.075 0.125	70	Standard:0.015 (Both sides)
MT-T-2X	0.075	0.075						
MT-T-4X	0.05	0.05						
MT-T-6X	0.032	0.05						

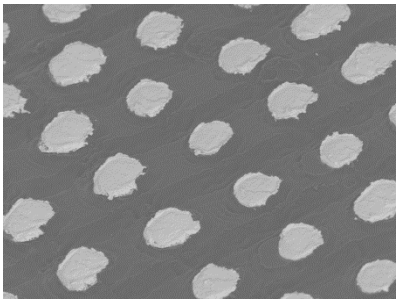
Operating Precautions

- In the case of solder electrodes, solder debris become deposited on the contact area after repeated use. Please clean the surface with a toothbrush; and remove the debris. Otherwise, resistance value will increase due to oxidation. When the solder debris is removed, the resistance value tends to recover.

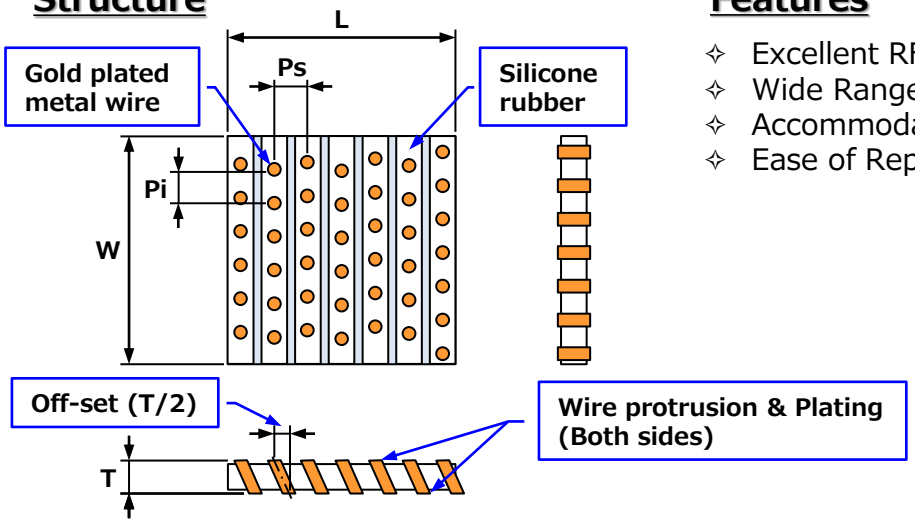
Shin-Etsu Inter-Connector “MT-P” series

- This sheet type connector for inspection is anisotropic conductive sheet type, which consists of gold plated metal wires embedded into silicone rubber. The conductive wires are designed in fine pitched matrix array.
- MT”P” Type will have added process of Au plating on the tip of the wire contacts. Excellent connection stability and high frequency characteristics.





Structure



Features

- Excellent RF characteristic
- Wide Range of Pitch and Sizes
- Accommodates various device types
- Ease of Repair and Replacement

Dimensions

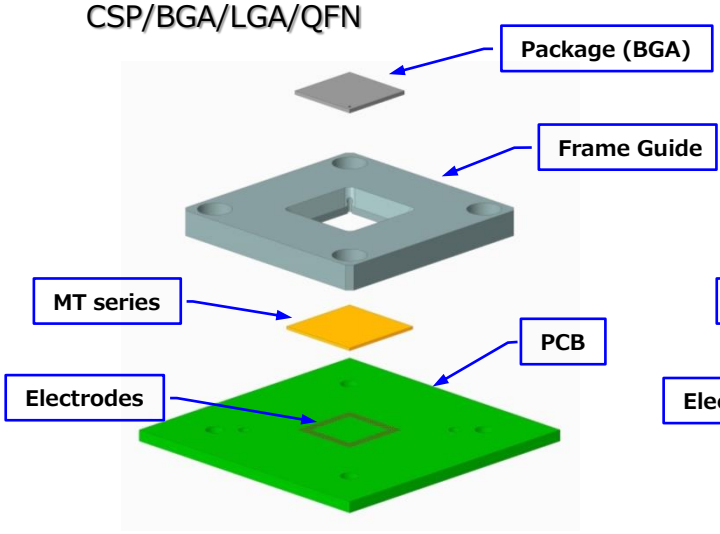
Type	Pi(P) (mm)	Ps (mm)	L (mm)	W (mm)	T (mm)	Off-set (mm)	Hardness	Protrusion of Wire (mm)
MT-P	0.1	0.1	5 to 50	5 to 50	(0.1) ・ 0.15 ・ 0.25	(0.05) ・ 0.075 ・ 0.125	70	Standard:0.02 (Both sides)
MT-P-2X	0.075	0.075						
MT-P-4X	0.05	0.05						

Operating Precautions

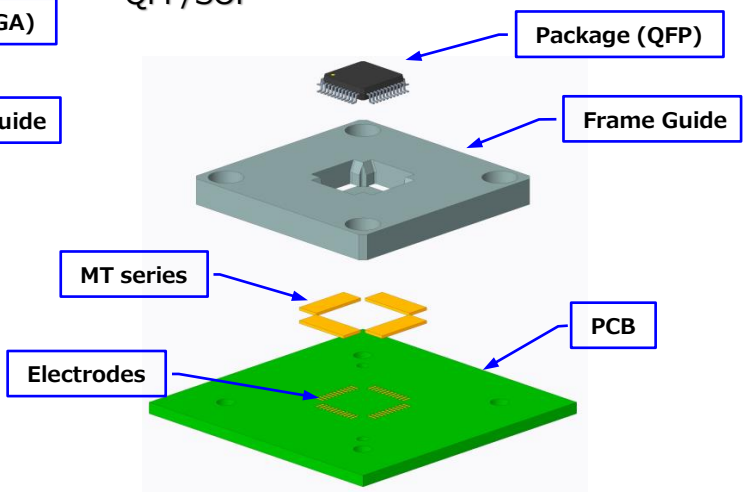
- In the case of solder electrodes, solder debris become deposited on the contact area after repeated use. Please clean the surface with a toothbrush; and remove the debris. Otherwise, resistance value will increase due to oxidation. When the solder debris is removed, the resistance value tends to recover.

Applications

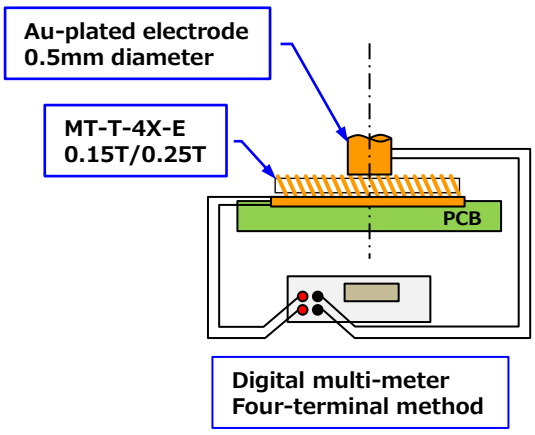
CSP/BGA/LGA/QFN

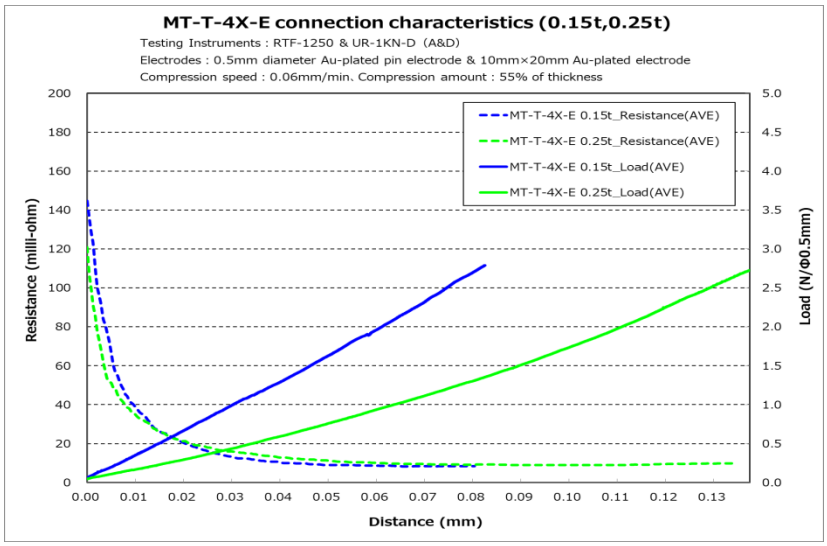


QFP/SOP

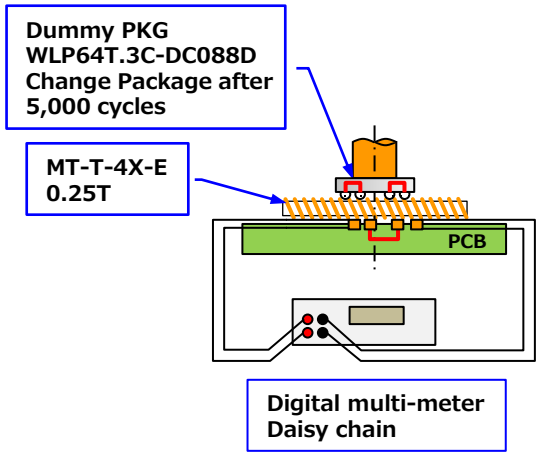


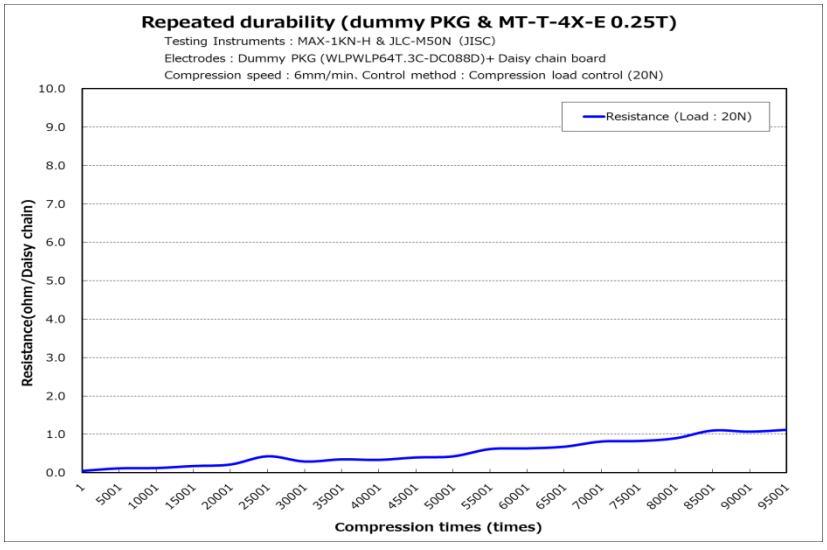
Connection Property



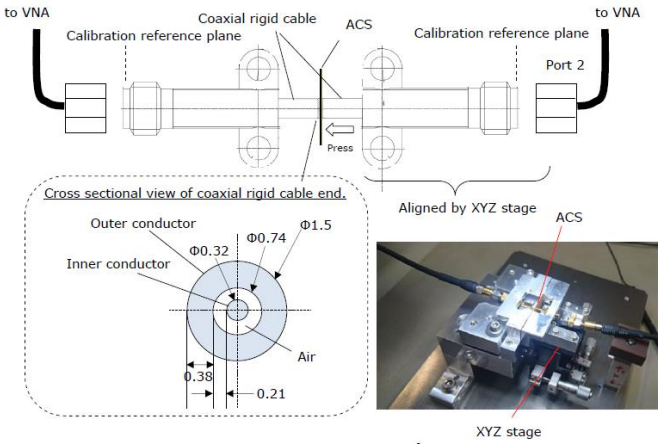


Repeated Durability Test





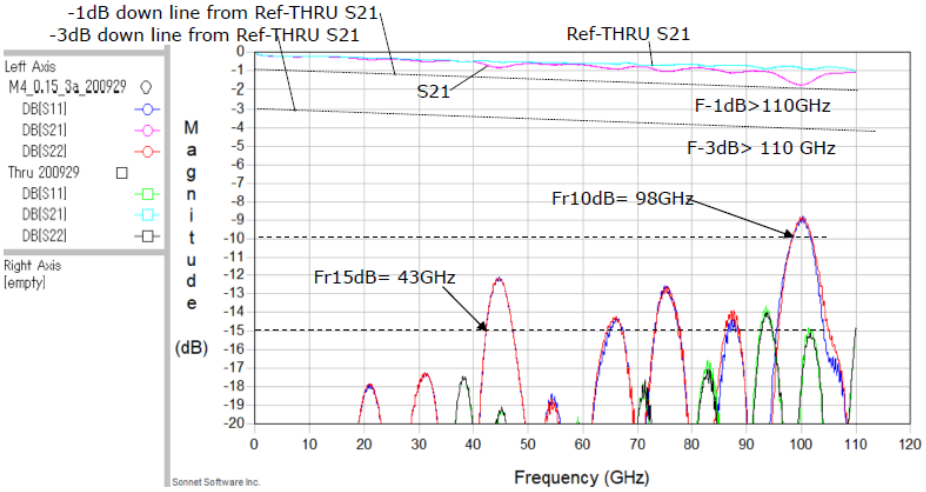
RF Performance

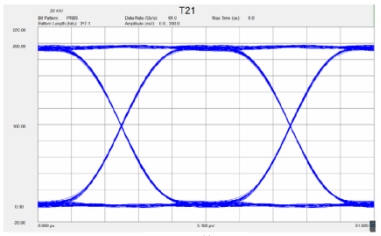


Evaluation Samples

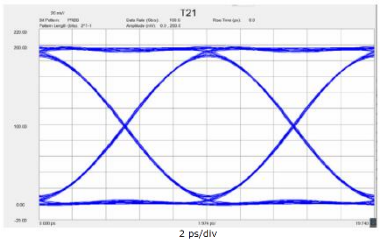
- MT-T-4X-E 0.15T
- MT-T-4X-E 0.25T

1. S-parameter and eye pattern for MT-T-4X-E 0.15T



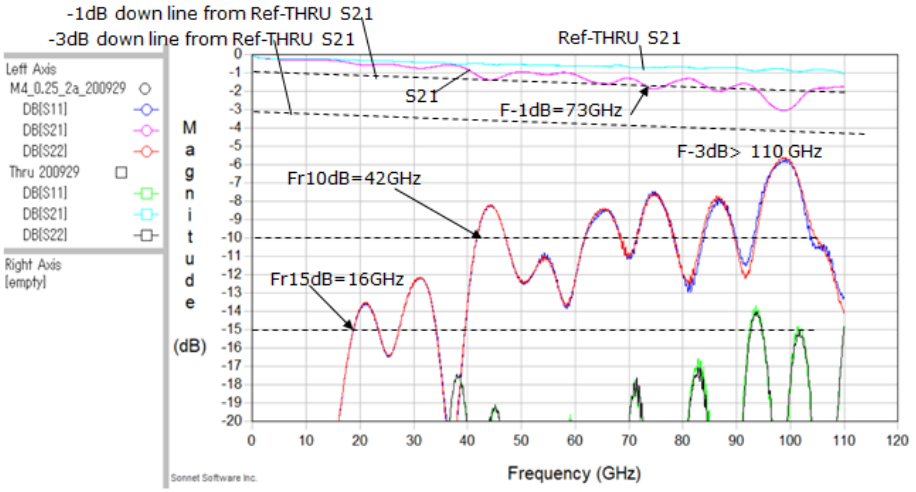


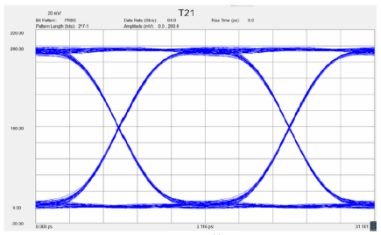
a) 64 Gb/s



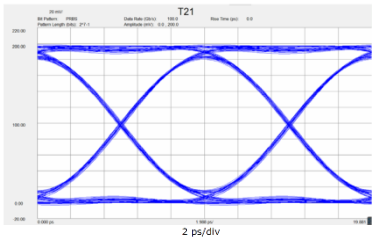
b) 100 Gb/s

2. S-parameter and eye pattern for MT-T-4X-E 0.25T





a) 64 Gb/s

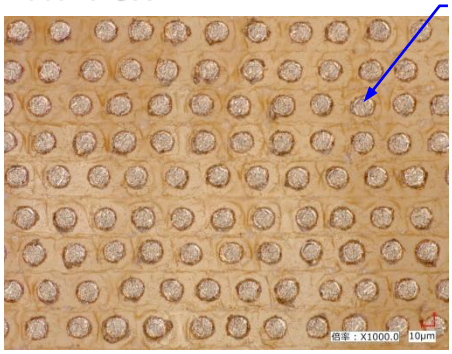


b) 100 Gb/s

NEW Development Product

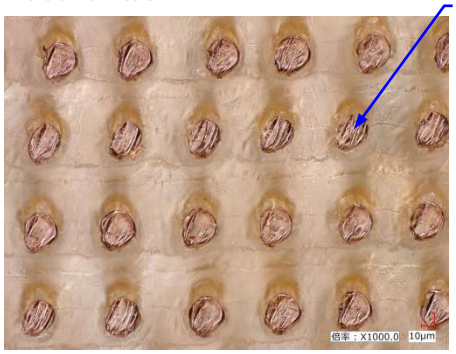
MT-T-9X: 30umPitch for small pitched devices

MT-T-9X



Comparison

MT-T-4X



Type	Pi(P) (mm)	Ps (mm)	L (mm)	W (mm)	T (mm)	Off-set (mm)	Hardness	Protrusion of Wire (mm)
MT-T-9X	0.03	0.03	10	10	0.15	0.075	70	Standard:0.015 (Both sides)
					0.25	0.125		

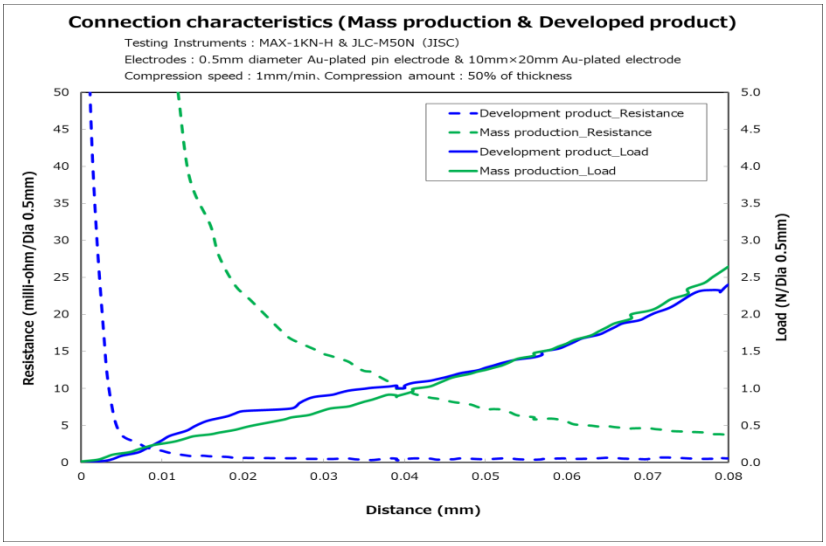
MT-LR : Using Low Resistivity Metal Wire

Graph

- Pitch(Pi x Ps):0.05Pi x 0.05Ps
- Size:10Lx10Wx0.25T

Features

- Low Resistance
- Good connectivity @ initial contact
- Excellent RF characteristic



Type	Pi(P) (mm)	Ps (mm)	L (mm)	W (mm)	T (mm)	Off-set (mm)	Hardness	Protrusion of Wire (mm)
MT-T-LR	0.1	0.1	5 to 20	5 to 20	0.15 0.25	0.075 0.125	70	Standard:0.015 (Both sides)
MT-T-LR-2X	0.075	0.075						
MT-T-LR-4X	0.05	0.05						
MT-T-LR-6X	0.04	0.04						